WordEase: Word Search Puzzle — Game Design & Build Flow

Oct 26, 2026 · @Claude User2

Overview

WordEase ek relaxing, large-print word search game hai jo React Native CLI mein banega, offline chalega, aur ads se kamayega. Player ek "World Trip" map par desh-desh ghoomta hai aur har stop ek word search level hai.

Item	Decision
Play Store title	WordEase: Word Search Puzzle
Short description	Large print word search for seniors. Big letters, no timer, play offline.
Platform	Android pehle (React Native CLI); iOS baad mein, same codebase
Target players	50+ adults (US, UK, Europe) + India (Hindi version)
Theme	World Trip: har world ek desh/region, har level ek city stop
Levels at launch	1,200 (20 worlds x 60), remote packs se 5,000+ tak
Kamai	Rewarded + capped interstitial + menu banner + Remove Ads IAP
Internet	Zaroori nahi; ads aur sync tab chalenge jab online ho

Competitors se alag kya hai:

- Text size control (Large se Huge) aur high-contrast mode
- Relax mode: no timer, no lives, no penalty
- Word meaning card: har mila word ka matlab
- Halke ads: pehle 3 levels mein koi ad nahi, interstitial par strict cap
- 6 bhashayein: English, Spanish, Portuguese, German, French, Hindi

- Apna World Trip map (kisi game ki copy nahi)

Success targets (pehle 90 din): Day-1 retention 35%+, Day-7 12%+, crash-free sessions 99.5%+, Play rating 4.5+.

Theme, art direction aur UI system

Look hoga "Paper Postcard World Trip": layered paper-cutout illustrations, halka paper grain, gol kinaare aur garam dhoop wale rang. Ye Candy Crush ke glossy candy look se bilkul alag hai, aur seniors ki aankhon par halka.

Art rules (no copy):

- Saare illustrations apne banwaye ya khud banaye hon; kisi game ke screenshot ya asset ko trace nahi karna
- Har world ka apna colour palette + ek generic landmark silhouette (jaise pyramid, pagoda, windmill) — kisi brand ya modern building ka logo nahi
- Free assets sirf clear license ke saath (jaise CC0); har asset ka license `assets/LICENSES.md` mein likhna

Colour tokens:

Token	Hex	Kaam
bg.cream	#FFF8EC	App background
ink	#1F2A37	Saara text (cream par contrast 13:1 se upar)
primary.ocean	#0E6E8C	Primary buttons, links
accent.coral	#F2785C	Current selection, Play button
coin.gold	#F5B700	Coins, stars
ok.green	#2E9B5E	Sahi word, success
found.1 to found.7	#E69F00, #56B4E9, #009E73, #F0E442, #0072B2, #D55E00, #CC79A7	Mile hue words ki highlight (Okabe-Ito colour-blind safe palette)

Typography:

Kahan	Font	Size (Normal / Large / Huge)
Grid letters	Atkinson Hyperlegible Bold (low-vision ke liye bana, OFL license)	28 / 34 / 42 sp
Word list chips	Atkinson Hyperlegible	20 / 24 / 28 sp
UI text, buttons	Nunito Bold (OFL)	18 / 21 / 24 sp
Hindi	Noto Sans Devanagari (OFL)	Upar jaisa

Accessibility rules:

- Har button kam se kam 48x48dp, hum 56dp rakhenge (grid cells ka size Core gameplay section mein); Android isi ko recommend karta hai (Android accessibility)
- Chhote text par contrast kam se kam 4.5:1; grid letters par 7:1 ka target
- System font scale follow karna + app ke andar Text Size slider
- High-contrast mode, colour-blind mode (found words par colour ke saath underline pattern), Reduce Motion toggle
- TalkBack ke liye har button ka label

Core components: PrimaryButton (56dp pill), IconButton (56dp), CoinPill, StarRow, PostcardModal, WordChip, GridCell, Toast, ProgressBar. Saare components `src/ui/` mein, ek theme file se rang aur size lenge.

Core gameplay

Player letter grid mein chhupe theme words dhoondhta hai: ungli se drag karke, ya pehla aur aakhri letter tap karke. Koi timer, life ya penalty nahi; galti par sirf halka shake.

Input ke do tareeke (Settings mein switch):

1. **Drag:** ungli rakho aur kheecho; selection 8 directions mein se sabse nazdeek wali line par snap hoti hai
2. **Tap-Tap:** pehla letter tap, phir aakhri letter tap. Haath kaanpne wale users ke liye zaroori

Cell size screen ke hisaab se banega: kam se kam 40dp (Normal), 48dp (Large), 56dp (Huge text). Agar bada grid chhote phone mein is size par fit na ho, toh grid par pinch-zoom aur pan khul jayega; tablet par poora grid ek saath dikhega. Drag mein halki galti maaf karne ke liye direction snapping 22.5° ke andar kaam karegi.

Difficulty ke saath kya badlega:

Levels	Grid	Words	Directions
1–30	6x6 – 7x7	4–6	Right, Down
31–120	7x7 – 8x8	6–8	+ Diagonal (neeche-right)
121–300	8x8 – 9x9	8–10	+ Ulta (left, up)
301–700	9x9 – 10x10	10–12	Saare 8
701+	10x10 – 12x12	12–14	Saare 8 + overlapping words

Huge text mode mein grid 10x10 se bada nahi hoga; generator us hisaab se chhota grid banayega.

Har level mein:

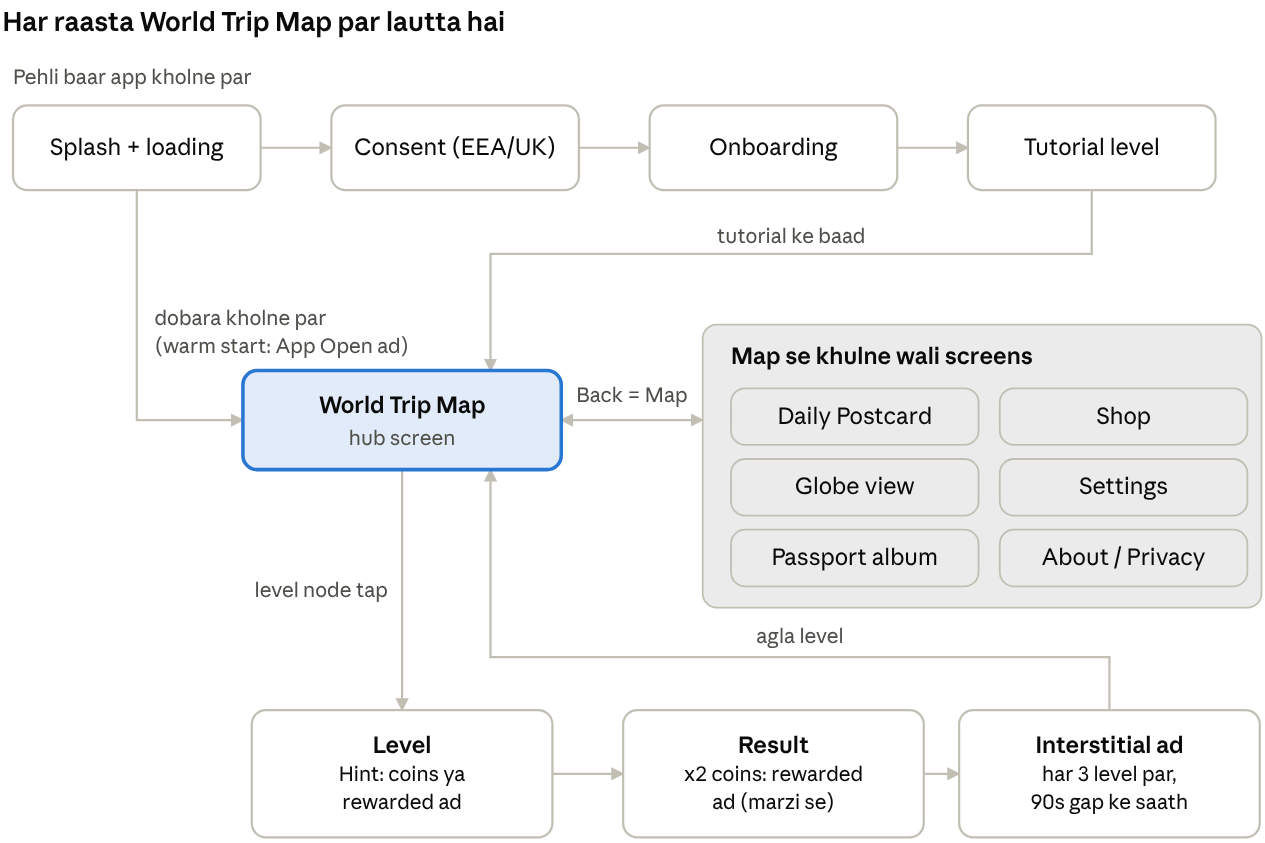
- Ek theme (jaise "Paris Cafe") aur us theme ke words, neeche chips mein
- Mila word: grid par rangeen capsule + chip par tick, chime aur halka haptic
- **Meaning card:** word milte hi neeche chhota card, matlab ke saath; 3 second mein khud band, tap karke khol sakte hain; Settings se off ho sakta hai
- **Bonus words:** theme ke bahar koi valid dictionary word mila toh +1 coin
- **Stars:** bina hint 3 star, 1 hint 2 star, 2+ hint 1 star (time se koi lena-dena nahi)

Hints (coins ya rewarded ad se):

Hint	Kya karta hai	Coins
Spotlight	Ek bache word ka pehla letter chamkata hai	25
Compass	Pehla letter + direction ka arrow	40
Reveal	Poora word dikha deta hai	60

Save: har mile word ke baad level ki state save hogi, taaki app band hone par bhi player wahi se shuru kare.

Screen-by-screen flow



WordEase screen flow · pehli baar, wapas aana, play loop

Naya player tutorial ke baad Map par aata hai; purana player seedha Map par. Har level ke baad player (zarurat ho toh ek interstitial ke saath) wapas Map par lautta hai. Daily Postcard ka puzzle bhi isi Level screen par khulta hai aur Calendar par lautta hai.

Screen	Kya hai	Ad
Splash + loading	Logo, fonts, audio aur level pack load	Warm start par App Open ad
Consent	Google UMP form, sirf EEA/UK/Switzerland	Koi nahi
Onboarding	2 steps: Text size chunna (live grid preview) + input mode (Drag ya Tap-Tap); Skip button	Koi nahi

Screen	Kya hai	Ad
Tutorial level	Level 1, haath ka pointer dikhata hai kaise drag karna hai, 3 words	Koi nahi
World Trip Map	Route map; upar coins, Daily, Settings; neeche banner	Banner
Globe view	Saare 20 worlds, band worlds baadalon ke neeche	Koi nahi
Level	Grid, word chips, Hint button, Pause; meaning card	Koi nahi (hint par rewarded sirf tap se)
Pause	Resume, Restart, Settings, Map	Koi nahi
Result	Sitare, coins, "x2 coins", Continue	Rewarded (x2); Continue ke baad interstitial (agar due)
Daily Postcard	Mahine ka calendar, streak, "Aaj khelo"	Koi nahi
Passport album	Har world ka stamp	Koi nahi
Shop	Coin packs, Remove Ads, Free coins	Rewarded (free coins)
Settings	Text size, input mode, music/SFX, haptics, meanings, high contrast, colour-blind, reduce motion, bhasha, Privacy options, Restore purchases	Koi nahi
About	Credits, WordNet notice, asset licenses	Koi nahi

Edge cases jo flow mein handle karne hain:

- Rewarded ad load nahi hua: button par "Ad abhi available nahi" + coins ka option; reward sirf ad poora hone ke callback par
- Ad ke beech app band ho gayi: reward nahi, aur agla interstitial counter reset nahi
- Internet nahi: game normal chalega; banner ki jagah fixed height reserved rahegi taaki layout na hile
- Phone call ya app background: level auto-pause, audio band

- Save file kharab: MMKV recovery + `schemaVersion` check; progress na mile toh bhi crash nahi
- Phone ki date peeche ki: Daily Postcard ka inaam usi din dobara nahi milega (claimed day save hota hai)
- Remove Ads ka payment pending: purchase "purchased" hone par hi ads band; Play ke rule ke mutabik purchase acknowledge karna, warna Google refund kar deta hai
- Phone par sirf portrait mode; tablet par portrait + landscape

World Trip map (level index screen)

Map ke do zoom levels honge: **Route view** (neeche se upar scroll hone wala raasta, har stop ek level) aur **Globe view** (poori duniya ka open-world style map, jahan locked desh baadalon ke neeche chhupe hain). Baadal hatna hi progress ka maza hai — ye feel open-world game maps jaisi hai, par design apna.

Route view:

- Neeche se upar scroll; ek world = 60 level nodes ek ghumavdaar dotted raaste par
- Background har world ka apna paper-cutout scene (jaise Egypt mein ret ke teele, Japan mein cherry trees)
- Player ka avatar ek chhota paper plane jo current level ke upar udta rehta hai
- Har 15th level "Landmark level": bada node, jeetne par passport stamp
- Screen khulte hi map khud current level par scroll karega
- Parallax: baadal aur patang dheere chalte hain (Reduce Motion on ho toh band)

Node states:

State	Look
Locked	Grey node, chhota lock
Current	Coral node, halki pulse animation, upar paper plane
Done	Ocean node + 1 se 3 sitare
Landmark	Bada gol stamp node, landmark icon

Globe view (pinch out ya globe button):

- 20 worlds ek stylised paper globe/map par; khule worlds rangeen, band worlds baadalon ke neeche
- World unlock hone par baadal udne ka animation + plane ka safar
- Kisi bhi khule world par tap karke seedha wahan jao

20 worlds (har world ki theme words bhi isi se):

#	World	#	World
1	Home Town (tutorial)	11	Australia Outback
2	Paris	12	Hawaii Islands
3	Italy	13	Mexico Fiesta
4	Greek Islands	14	Peru Mountains
5	Egypt	15	Brazil Rainforest
6	Kenya Safari	16	Canada Lakes
7	India	17	New York
8	Thailand	18	Iceland Lights
9	Japan	19	Scotland Highlands
10	Great Wall	20	Antarctica

Passport album: har world khatam karne par ek stamp; album screen mein saare stamps — ye long-term goal deta hai.

Performance plan:

- Map ko "chunks" mein todna: ek chunk = ek world; sirf screen ke aas-paas ke 2 chunks render honge (FlashList virtualization)
- Background WebP mein, do resolution (normal aur low-RAM phones ke liye)
- Raasta aur sitare Skia canvas par; level nodes asli Pressable buttons, taaki TalkBack unhe padh sake
- Target: mid-range Android par 60 FPS, map open hone mein 1 second se kam

Levels: generator, difficulty aur word lists

Launch par 1,200 levels (20 worlds x 60) honge. Ye ek Node.js script se build-time par generate, check aur JSON packs mein save honge; app sirf ready levels load karega. Baad mein naye worlds remote packs se aayenge, bina app update ke.

Generator kaise kaam karega:

1. Level ki theme, word list, grid size aur allowed directions lo
2. Words ko lambe se chhote order mein lagao
3. Har word ke liye saari possible jagah + direction ki list banao, random shuffle karo, pehli fit hone wali jagah par rakho
4. Koi word fit na ho toh pichhla word hata kar dobara try karo (backtracking); 200 tries ke baad naye seed se restart (algorithm reference)
5. Khaali cells ko bhasha ke letter-frequency ke hisaab se random letters se bharo
6. Seeded random (jaise mulberry32) use karo, taaki same seed se hamesha same puzzle bane

Har level ki automatic checks (CI mein):

- [ ] Har theme word grid mein theek ek hi baar ho (galti se doosri copy nahi)
- [ ] Kisi bhi direction mein koi gaali ya offensive word na bane (blocklist scan)
- [ ] Diagonal words ka hissa level ke hisaab se (minimum 20% jab diagonal allowed ho)
- [ ] Bonus words ki list nikaal kar level JSON mein save
- [ ] Har word ki meaning available ho

Har world ki difficulty rhythm (sawtooth):

Levels (world ke andar)	Feel
1–5	Aasaan warm-up, naye world ka swagat
6–14	Dheere-dheere mushkil
15	Landmark level (peak)
16–17	Breather: aasaan, inaam jaisa
18–44	Phir chadhaav, landmark 30 aur 45 par
46–59	Sabse mushkil hissa
60	World finale + passport stamp

Difficulty score = grid size + words ki ginti + directions + lambe words + overlaps + decoy letters. Upar ke levels mein khaali cells mein theme words ke milte-julte tukde (decoys) daale jayenge, taaki dhoondhna thoda mushkil ho.

Word lists aur licenses:

Kya	Source	License
Spelling check + bonus words	ENABLE word list	Public domain
English meanings	WordNet 3.0	Commercial use allowed; copyright notice app ke About screen mein dikhana zaroori
Theme word lists	Khud curate kiye (AI se draft, phir haath se review)	Apne
Doosri bhashayein	Har bhasha ke liye curated list + native speaker review	Apne

Theme lists family-friendly hongi: koi gaali, politics, religion ya brand naam nahi.

Content volume: 1,200 levels x ~9 words = lagbhag 10,000 word slots; ek theme ke words kai levels mein dohraaye ja sakte hain, isliye ~600 themes kaafi hongi.

Players sab levels khatam kar dein toh: "Classic Play" mode khul jayega, jo kisi bhi theme se naye puzzles turant generate karta hai, taaki game kabhi khatam na ho.

Daily puzzle, streaks aur coin economy

Coins hi game ki ek currency hongi; inse sirf hints aur cosmetic cheezein aayengi, koi level coins se lock nahi hoga. Target: bina ad dekhe player roz 2–4 hints afford kar sake.

Daily Postcard (daily puzzle):

- Har din ek special puzzle, phone ki local date se seed hota hai
- Calendar screen par mahine ke din; har pura din ek postcard collect karta hai
- Streak: lagatar din; har hafte 1 free "streak freeze" (ek din chhoot jaye toh streak nahi tootegi)
- Inaam: roz 50 coins; din 3 par 100, din 7 par 300 + naya plane skin

Coins kamana:

Kaise	Coins
Level jeetna (1 / 2 / 3 star)	10 / 15 / 20
Bonus word	1 har word
Daily Postcard	50
World khatam	150
Result screen par "x2 coins" (rewarded ad)	Level ke coins double
Shop mein "Free coins" (rewarded ad, din mein max 5 baar)	50
Naye player ko shuru mein	200

Coins kharchna:

Kya	Coins
Spotlight / Compass / Reveal hint	25 / 40 / 60
Plane skins (map avatar)	500 – 2,000
Map themes (jaise night mode map)	1,500

In-app purchases (Google Play Billing):

Product	Price (USD)	Kya milega
Remove Ads	3.99	Banner, interstitial, app-open ads hamesha ke liye band; rewarded ads optional rehte hain
Coin pack S	0.99	500 coins
Coin pack L	2.99	2,000 coins

India ke liye Play Console mein local pricing (jaise Remove Ads ₹149) set karni hai.

Reminders: din mein sirf ek notification, player ke chune time par, aur sirf opt-in ke baad. Notification permission level 5 ke baad maangni hai, app khulte hi nahi.

Ads plan

Chhe jagah ads honge, saare AdMob ke rules ke andar: gameplay screen par kabhi koi ad nahi, pehle 3 levels mein koi ad nahi, aur har full-screen ad ke beech kam se kam 90 second ka gap. Saari limits Remote Config se badli ja sakengi.

#	Format	Kahan	Kab / limit
1	Rewarded	Hint button par (coins kam hon toh "Watch ad for free hint")	Sirf player ke tap par; din mein max 10
2	Rewarded	Result screen par "x2 Coins"	Player ki marzi; har level par ek baar
3	Rewarded	Shop mein "Free 50 coins"	Din mein max 5
4	Interstitial	Result screen par "Continue" dabane ke baad, map par lautne se pehle	Level 4 se shuru; har 3 levels mein ek; 90 second gap; landmark level ke baad nahi
5	App Open	App background se wapas aane par (warm start), loading screen ke upar	Install ke 2 din baad se; 4 ghante mein max 1; agar 90 second ke andar koi aur full-screen ad dikha ho toh nahi
6	Adaptive banner	Sirf Map screen ke sabse neeche	Neeche ke 80dp mein koi level node nahi; banner aur map ke beech 12dp gap

AdMob rules jo tootne nahi chahiye:

- Interstitial app khulne ya band hone par nahi; har tap ke baad nahi (do user actions par max ek); ek interstitial ke turant baad doosra nahi; sirf levels jaise logical break par (AdMob interstitial guidance)
- Banner gameplay screen par nahi aur buttons ke bilkul paas nahi, warna galti se click aur account ban ka khatra (AdMob banner guidance)
- App open ad loading screen ke upar dikhe, aur uske turant pehle ya baad koi doosra ad nahi (App open guidance)
- Designed for Families program wale apps app open ads use nahi kar sakte, isliye Play Console mein target audience 18+ rakhni hai
- Rewarded ad hamesha player ke tap se shuru ho, aur reward ad poora hone par hi mile

Consent (GDPR / UMP): pehli baar app khulne par, map dikhne se pehle, Google UMP consent form (sirf EEA, UK, Switzerland mein dikhega). Consent ke baad hi ads SDK initialize hoga. Settings mein "Privacy options" button jisse player consent badal sake.

Mediation (launch ke 1 mahine baad): AdMob mediation mein bidding partners jodna: InMobi (India mein strong), AppLovin, Meta Audience Network, Unity Ads, Liftoff. Har adapter Android Gradle mein alag add hota hai.

Safety checklist:

- [ ] Development mein sirf test ad IDs; apna phone test device ke roop mein register
- [ ] Kabhi apne ads par click nahi, dost-rishtedaron ko bhi nahi
- [ ] `app-ads.txt` apni website par
- [ ] Har ad impression ki revenue (paid event) Firebase Analytics mein log
- [ ] Remove Ads kharidne wale user ke liye banner, interstitial aur app open poori tarah band

Remote Config keys:

Key	Default
ads_first_interstitial_level	4
ads_interstitial_every_n_levels	3
ads_min_gap_seconds	90
ads_app_open_enabled	true

Key	Default
ads_app_open_min_hours	4
ads_banner_on_map	true
ads_daily_rewarded_hint_cap	10

Music, sound effects aur haptics

Audio ke liye Software Mansion ki **react-native-audio-api** use hogi: iska core C++ mein hai, Android par Oboe use karta hai, aur ek saath kai sounds kam latency par chala sakta hai (docs). Music aur SFX ke alag volume control (do alag gain nodes) honge.

Music:

- 6 shaant loops (60–120 second), har ek mein region ki halki jhalak: Paris mein accordion, India mein sitar jaisi dhun, Japan mein koto — par sab original
- Music sirf commission karwa ke ya commercial license ke saath kharid ke; license file `assets/LICENSES.md` mein
- Format: OGG ya M4A, ~128 kbps; map aur level music ke beech crossfade
- Full-screen ad khulte hi game ka saara audio pause, ad band hone par resume
- App background mein jaate hi audio pause
- Default: music 50%, SFX 80%; Settings mein dono ke alag slider aur on/off

Sound effects:

Event	Sound
Letter par ungli	Halka wood click
Drag mein har naya letter	Tick, har letter par pitch thodi upar
Word mila	Garam chime chord
Galat selection	Naram thud (koi buzzer nahi)
Level complete	2 second ki chhoti fanfare
Sitare (1 se 3)	Har sitare par pop
Coins	Sikkon ki khanak

Event	Sound
Map par plane	Whoosh
Passport stamp	Stamp ki thump
Button tap	Soft click

Saare SFX 0.1–1.5 second ke, WAV mein, app start par memory mein preload.

Haptics (react-native-haptic-feedback, TurboModule support ke saath):

Event	Haptic
Drag mein naya letter	Selection tick
Word mila	Success notification
Button tap	Light impact
Level complete	Medium impact

Settings mein Haptics on/off. Reduce Motion on ho toh celebration animations chhoti ho jayengi, par sound rahega.

Tech stack (React Native CLI)

React Native 0.84+ CLI + TypeScript, sirf New Architecture par. 0.82 se purani architecture hat chuki hai, aur 0.84 se Hermes V1 default engine hai (RN 2026 overview). Har library New Architecture support wali chuni gayi hai.

Kaam	Library	Kyun / note
Grid, map path, effects	@shopify/react-native-skia 2.x	GPU drawing; RN 0.79+ aur React 19 chahiye (Skia guide)
Animation	react-native-reanimated 4 + react-native-worklets	UI thread par 60/120 FPS; babel mein `react-native-worklets/plugin` sabse aakhir mein (Reanimated 4 guide)

Kaam	Library	Kyun / note
Gestures (drag select)	react-native-gesture-handler 2.x	Stable; v3 (May 2026) React Navigation 8 ke saath lenge (version note)
Navigation	@react-navigation/native 7 + native-stack	v8 abhi alpha mein hai, stable aane par upgrade (v8 release)
Map list virtualization	@shopify/flash-list v2	New Architecture only, size estimate ki zaroorat nahi (FlashList v2)
Local save	react-native-mmkv v4 + react-native-nitro-modules	Synchronous, AsyncStorage se ~30x tez; sirf 64-bit ABIs (MMKV v4)
State	zustand + MMKV persist	Chhota, simple, boilerplate kam
Ads + consent	react-native-google-mobile-ads	AdMob + UMP consent ek hi package mein (library)
In-app purchase	react-native-iap (OpenIAP, Nitro)	Google Play Billing 8+, RN 0.79+ chahiye (react-native-iap)
Analytics, crashes, remote config	@react-native-firebase v26 (app, analytics, crashlytics, remote-config)	v26 se New Architecture zaroori (migration v26)
Audio	react-native-audio-api	Low latency SFX + music
Haptics	react-native-haptic-feedback	TurboModule support
Celebrations	lottie-react-native (chhote JSON) ya Skia particles	Confetti, stamp animation
Splash screen	react-native-bootsplash	Native splash, app open ad ke neeche loading screen
Bhashayein	i18next + react-i18next	Har bhasha ki JSON file
Testing	Jest, React Native Testing Library, Maestro (E2E)	Neeche Testing section

Setup se pehle verify karna (versions badalte rehte hain): local notifications library (jaise Notifee), in-app review library, aur Hindi grapheme splitting (Hermes mein `Intl.Segmenter` ya `graphemer` package) — teeno ka New Architecture + RN 0.84 support install se pehle check karna.

Android build settings:

- `newArchEnabled=true`, Hermes on
- `reactNativeArchitectures=arm64-v8a,x86_64` (MMKV v4 32-bit support nahi karta)
- Play Console ki current target API requirement follow karni hai; 2026 mein naye apps ke liye Android 15 (API 35) ya usse upar (publishing guide)
- Release AAB, R8/ProGuard on, Play App Signing

App architecture: folders, state aur data

Code chaar hisson mein bata hoga: screens (UI), game engine (pure TypeScript, bina React ke, taaki test ho sake), services (ads, IAP, audio, analytics) aur store (zustand + MMKV). Level generator ka code app aur build script dono share karenge.

Folder structure:

```
src/
  app/          App.tsx, providers, navigation
  screens/      Splash, Consent, Onboarding, Map, Globe, Level, Result,
                Daily, Album, Shop, Settings, About
  game/
    engine/     selection.ts (8-direction snap), validate.ts, hints.ts,
scoring.ts
    generator/  prng.ts, placeWords.ts, fill.ts, checks.ts
    render/     GridCanvas.tsx (Skia), Highlights.tsx, LetterCell.tsx
  map/          RouteMap.tsx, WorldChunk.tsx, LevelNode.tsx, Globe.tsx
  services/
    ads/        AdsManager.ts (caps + timing), consent.ts, adUnits.ts
    iap/        store.ts
    audio/      AudioManager.ts
    analytics/  events.ts
    remote/     remoteConfig.ts
  store/        progress.ts, wallet.ts, settings.ts, daily.ts, adState.ts
  ui/           theme.ts, PrimaryButton, CoinPill, PostcardModal, ...
  i18n/         en.json, es.json, pt.json, de.json, fr.json, hi.json
assets/
```

```
  levels/en/    world-01.json ... world-20.json
  images/  audio/  fonts/  lottie/  LICENSES.md
tools/
  generate-levels.ts, validate-levels.ts, blocklist/
```

Main data models:

```
type Dir = 'E' | 'S' | 'SE' | 'W' | 'N' | 'NW' | 'NE' | 'SW';

type LevelDef = {
  id: number; world: number; theme: string;
  rows: number; cols: number; grid: string[];   // har row ek string
  words: { w: string; start: [number, number]; dir: Dir }[];
  bonus: string[]; seed: number; difficulty: number;
};

type Progress = {
  currentLevel: number;
  stars: Record<number, 1 | 2 | 3>;
  inProgress?: { levelId: number; found: string[]; hintsUsed: number };
};

type Wallet = { coins: number; adFree: boolean };

type Settings = {
  textScale: 1 | 1.25 | 1.5; inputMode: 'drag' | 'tap';
  music: number; sfx: number; haptics: boolean; meanings: boolean;
  highContrast: boolean; colorBlind: boolean; reduceMotion: boolean; lang:
string;
};

type AdState = {
  levelsSinceInterstitial: number; lastFullscreenAt: number;
  lastAppOpenAt: number; rewardedHintsToday: number; dayKey: string;
};

type Daily = { streak: number; lastDay: string; freezes: number; done:
Record<string, true> };
```

Save aur load:

- Do MMKV instances: `progress` (progress, wallet, daily, adState) aur `settings`

- Har mile word par `inProgress` save; level khatam hone par progress + coins ek saath save
- `schemaVersion` key, taaki aage data badalne par purana save migrate ho sake
- Android Auto Backup on (phone badalne par progress wapas); cross-device cloud save Phase 2 mein

Performance budget:

Cheez	Target
Cold start (mid-range Android)	2.5 second se kam
Level khulna	300 ms se kam
Drag ke dauran	60 FPS, frame drop nahi
Download size (pehla install)	60 MB se kam; baad ke worlds ki images Phase 2 mein download
Memory	2 GB RAM phone par crash nahi

Bhashayein (localization)

Launch English mein, phir month 3 tak 5 aur bhashayein. Har bhasha ke apne word lists, apni store listing aur apna keyword-wala title hoga; UI language aur Puzzle language Settings mein alag-alag chuni ja sakengi.

Bhasha	Play Store title (30 characters tak)	Note
English	WordEase: Word Search Puzzle	Launch
Spanish	WordEase: Sopa de Letras	"Sopa de letras" wahan ka asli search word
Portuguese (Brazil)	WordEase: Caça-Palavras	Brazil bada market
German	WordEase: Wortsuche	High ad rates
French	WordEase: Mots Mêlés	

Bhasha	Play Store title (30 characters tak)	Note
Hindi	WordEase: शब्द खोज	India market

Hindi grid ke khaas rules:

- Ek cell = ek akshar (jaise "कि", "मा", "क्ष"), ek Unicode code point nahi
- Akshar todne ke liye grapheme splitting zaroori hai jo Unicode 15.1+ ke Indic conjunct rule ko support kare, warna "क्ष" do cells mein toot jayega
- Khaali cells ko common aksharon ki frequency list se bharna
- Devanagari font size Latin se ~10% bada, taaki matra saaf dikhe

Process:

- [ ] Saare UI strings `i18n/*.json` mein, code mein koi hard-coded text nahi
- [ ] Har bhasha ki theme word list native speaker se review
- [ ] Har bhasha ke liye alag blocklist (gaali scan)
- [ ] Har store listing ke screenshots usi bhasha mein
- [ ] Lambe German/French words ke liye buttons par text wrap test

Analytics, Remote Config aur A/B tests

Firebase Analytics + Crashlytics + Remote Config se teen sawaalon ka jawab milega: log kahan chhod rahe hain, kaunse levels zyada mushkil hain, aur kitne ads bina retention giraye chal sakte hain. Koi naam, email ya personal data collect nahi hoga.

Events:

Event	Parameters
app_open	type (cold / warm)
consent_result	status
tutorial_step	step
level_start	level_id, world
word_found	level_id, word_length, direction
hint_used	type, source (coins / ad)

Event	Parameters
level_complete	level_id, stars, seconds, hints
level_quit	level_id, words_found
ad_impression	format, placement, value (paid event)
rewarded_complete	placement
iap_purchase	product_id
daily_complete	streak
world_unlock	world
setting_change	key, value

Roz dekhne wale numbers: Day-1/7/30 retention, levels per daily user, ad revenue per daily user, rewarded opt-in rate, crash-free sessions, aur har level ka quit rate.

Mushkil levels ko theek karna: jis level ka quit rate baaki levels se kaafi zyada ho, uska aasaan version generate karke remote pack se replace karna — bina app update ke.

A/B tests (ek baar mein ek, har test 2 hafte):

1. Interstitial har 3 vs har 4 levels
2. Pehla interstitial level 4 vs level 6
3. Hint prices: abhi wale vs 20% sasti
4. Map par banner: on vs off
5. Meaning card default: on vs off

Har test mein revenue ke saath Day-7 retention bhi dekhna hai; retention gire toh zyada revenue wala version bhi nahi lena.

Play Console Data safety form mein: Advertising ID, app interactions (analytics), crash logs, purchase history (IAP) declare karna.

Testing plan

Testing paanch layers mein hogi: engine unit tests, generator ke 10,000-seed tests, UI tests, asli phones par E2E, aur 14 din ka closed test. Release tabhi jab neeche ke saare gates pass hon.

1. Unit tests (Jest):

- Selection snapping: har 8 direction, ulta drag, aadha drag, grid ke kinare
- Word match: aage aur ulta dono, ek letter wala galat match nahi
- Hints: sahi word chuna, coins kate, bache words khatam hone par button band
- Stars aur coins ka hisaab
- AdsManager (nakli clock ke saath): pehle 3 levels mein ad nahi, har 3 level par ek, 90 second gap, app open ke turant baad interstitial nahi, Remove Ads ke baad koi ad nahi
- Daily streak: aadhi raat, timezone badalna, phone ki date peeche karna, streak freeze
- Save migration: purane `schemaVersion` ka data naye mein

2. Generator tests (CI mein, fast-check jaisa property testing):

- 10,000 random seeds par: har word ek hi baar, saare cells bhare, sirf allowed directions, koi blocklisted word nahi
- Saare 1,200 shipped levels dobara validate; ek bhi fail toh build fail

3. UI tests (React Native Testing Library): buttons, Settings toggles, Text Size par layout, Hindi text rendering.

4. End-to-end (Maestro, asli phone par):

1. Pehli baar app → consent → tutorial → level 1 jeetna → result → map
2. Hint button → rewarded test ad → hint milna
3. Level ke beech app band karna → dobara kholna → wahi se shuru
4. Bina internet poora level khelna
5. Remove Ads kharidna (Play license tester) → reinstall → purchase restore
6. Bhasha badalna, Text Size Huge, Tap-Tap input mode
7. 10 levels lagatar: interstitials sahi jagah aur sahi ginti mein

5. Device matrix:

Device type	Kyun
2 GB RAM low-end Android	Memory aur map scroll
Mid-range Android 13–14	Asli user ka aam phone
Naya flagship Android 15+	Edge-to-edge, naya OS
7–10 inch tablet	Seniors tablet bhi use karte hain

Device type	Kyun
System font size "Largest"	Layout tootna nahi chahiye
TalkBack on	Accessibility

Ads aur IAP testing: AdMob Ad Inspector se har ad unit check; UMP consent ko EEA debug geography se test; Play Console license testers se test purchases, refund aur restore.

Release gates:

- [ ] Saare unit + generator tests pass
- [ ] Maestro ke saare 7 flows pass
- [ ] Play Console pre-launch report mein koi crash nahi
- [ ] Accessibility Scanner mein koi chhota touch target nahi
- [ ] Drag ke dauran 60 FPS (Perfetto / Android profiler)
- [ ] Closed test: 20+ testers, 14 din lagatar opt-in, kam se kam 5 asli 50+ umar ke users se feedback
- [ ] Crash-free sessions 99.5%+ closed test mein

Play Store launch checklist aur ASO

Launch se ~3 hafte pehle closed test shuru karna hai, kyunki naye personal account ko production se pehle 12+ testers ke saath lagatar 14 din test chalana padta hai (12-tester rule).

Play Console checklist:

- [ ] Developer account + identity verification
- [ ] Privacy policy URL (website par, PDF nahi)
- [ ] Data safety form (Analytics section dekho)
- [ ] Content rating questionnaire (IARC)
- [ ] Target audience 18+ (Families program se bahar, warna app open ads nahi chalenge)
- [ ] "Contains ads" declaration on
- [ ] `app-ads.txt` website par, AdMob se linked
- [ ] In-app products: Remove Ads, coin packs (local prices ke saath)
- [ ] Release AAB, Play App Signing

- [ ] Closed testing track: 20+ testers, 14 din

Store listing (har bhasha mein):

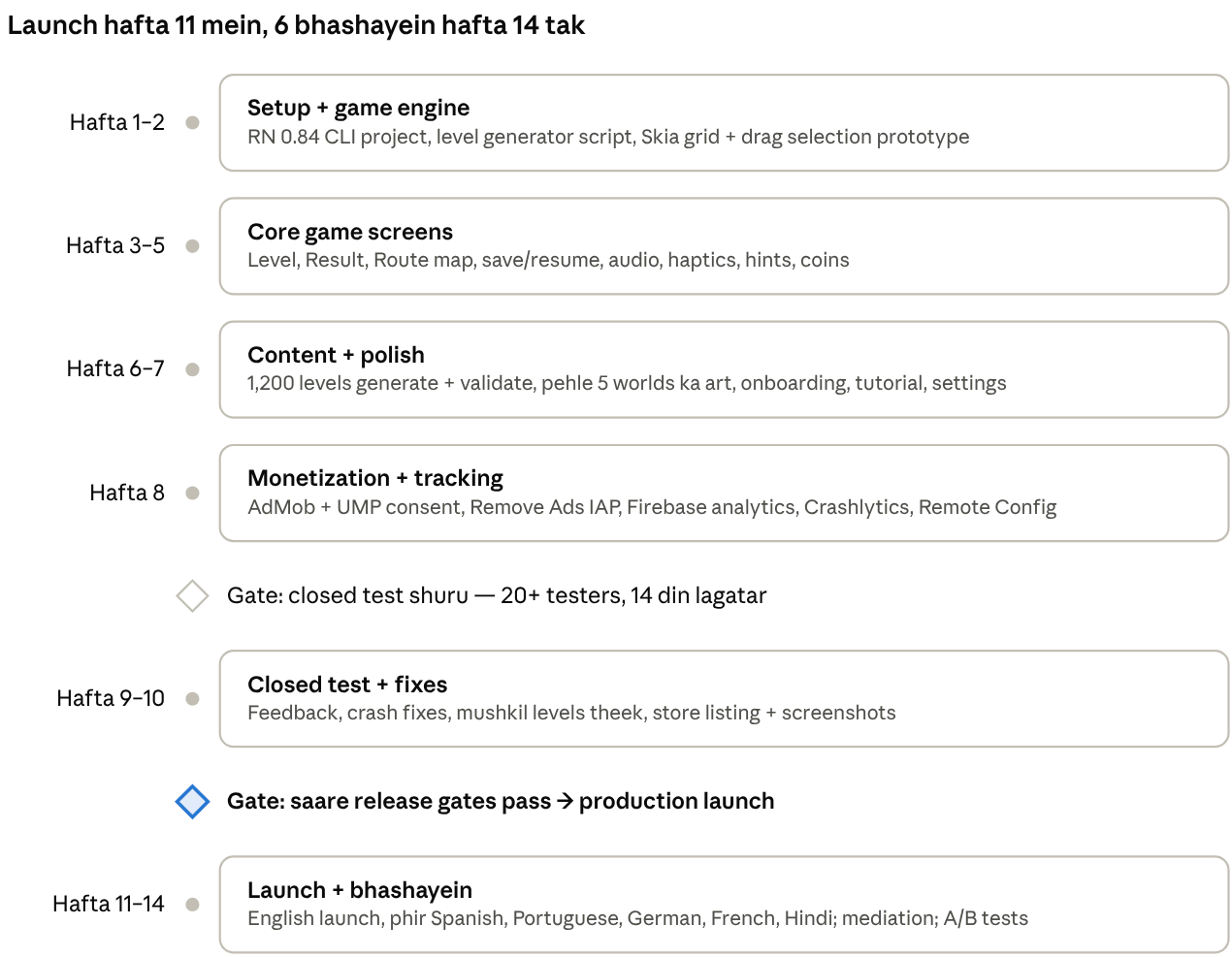
Cheez	Plan
Title	WordEase: Word Search Puzzle
Short description	Large print word search for seniors. Big letters, no timer, play offline.
Long description	~1,800 characters; keywords naturally: large print, big letters, relaxing, offline, no timer, word meanings, daily puzzle, world trip
Icon (512x512)	Bade "W" letter wala paper-postcard tile, ek nazar mein padha jaye
Feature graphic (1024x500)	Map + grid ek saath
Screenshots	10–12; pehle 3 mein bade letters, map aur "no timer" saaf dikhe
Promo video	20–30 second gameplay

1M+ installs wale apps ke listings mein median 12 screenshots aur ~1,807 characters ki description hoti hai, aur 28% ke paas promo video hota hai (listing study).

Review maangna: Google Play In-App Review prompt sirf 3-star jeet ke baad, level 10 ke baad, aur 30 din mein max ek baar. Kabhi bhi "5 star do toh coins milenge" jaisa inaam nahi (policy ke khilaaf).

Launch ke baad har hafte: reviews ka jawab, quit-rate wale levels theek karna, ek naya A/B test, aur keyword ranking dekh kar description sudharna.

Build roadmap



Build roadmap · 6 phases, 2 gates

Akele developer ke liye ~11 hafte mein Play Store launch, aur 14ve hafte tak 6 bhashayein. Sabse lamba hissa core game screens (3 hafte) hai; closed test ke 14 din ko kam nahi kiya ja sakta, isliye hafta 8 ke end tak build testers ke haath mein honi chahiye.

Pehle hafte ke kaam:

- [ ] `npx @react-native-community/cli init WordEase` (TypeScript), New Architecture + Hermes on
- [ ] Skia, Reanimated 4 + worklets, Gesture Handler, MMKV v4 install karke ek blank screen par test
- [ ] `tools/generate-levels.ts` ka pehla version: 1 theme, 10 levels
- [ ] Skia par 8x8 grid + drag selection + capsule highlight ka prototype
- [ ] Play Console developer account banana (verification mein kuch din lagte hain)

Sources

- Android accessibility guidelines
- AdMob: Disallowed interstitial implementations
- AdMob: Banner ad guidance
- AdMob: App open ad guidance
- react-native-google-mobile-ads
- React Native 2026 features (New Architecture only, Hermes V1)
- React Native Skia 2026 guide
- Reanimated 4 guide
- Reanimated / Gesture Handler versions
- React Navigation 8 alpha release
- FlashList v2
- react-native-mmkv v4
- react-native-iap
- React Native Firebase v26 migration
- React Native Audio API docs
- Word search generator algorithm
- ENABLE word list (public domain)
- WordNet license
- Google Play 12-tester rule
- Play Store listing study
- Publishing guide (target API)